

Part B: Take-Home Test

Examiners: Mr MJ Legodi
Moderator: Mr RQ Odendaal

Release Date: 21 April 2021 at 12:40
Due Date: 21 April 2021 at 14:40

Please read and understand the following.

1. All test and examination regulations of the University of Pretoria apply.
2. This is an OPEN BOOK test. You may use any source available to you, like your textbook and lecture notes, to assist in answering the questions. However, you **MAY NOT** receive (or provide) help in any way whatsoever from (or to) any other person. This includes the lecturer, tutor, a friend and family members.
3. Answer the questions **in your own handwriting** on separate paper, lined or unlined.
Take quality photographs of your answer sheets and scan them as a PDF document. Use apps like **CamScanner** to convert the photographs to PDF.
Submit your answer sheets, via ClickUP, in the “Semester Test 1 (Part B, non-WebAssign) Submissions” in the “Semester Test 1 Folder” under the “TESTS & EXAMS” headline. **Submit before 14:40.**
There will be a queue during submission. You may not get your submission receipt immediately! To accommodate this, we give you a grace period of 20 minutes only, between 14:40 and 15:00 to make your submission.
YOU HAVE BEEN WARNED: Late submissions will not be accepted!
4. Number and label your submissions as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
5. Provide neatly labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct numbers of significant figures, units, etc. Failure to do so will result in marks being deducted.
6. Your answer needs to be written in a systematic and logical fashion. Map out the steps you will take to reach your goal i.e. solve the problem in your mind first! You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not necessarily mean that you will receive full marks!
7. Units must be included when numbers are substituted; else, marks will be deducted.
8. You will need to look up constants and typical values for parameters in your calculations. Use your textbook or any credible source.
9. If any misconduct with respect to this test is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.

DECLARATION OF ORIGINALITY

UNIVERSITY OF PRETORIA - Department of Physics

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I understand what plagiarism is and am aware of the University's policy in this regard. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This statement is a reminder of the Code of Conduct that you have already signed.

TOTAL POINTS (Part B): 43

Question 1 [17]

- 1.1 A surveyor is measuring distances in a level yard near a road. A fence post is due north from the surveyor and is 18.1 m away from him. A street sign is to the right of the fence post from the surveyor's view, and is a distance of 24.4 m from him. The surveyor measures the angle between the fence post and the street sign to be 22.0° . (**HINT:** The surveyor, the fence post and the street sign can be viewed as the vertices of a triangle. None of the angles in this triangle is a right angle.)

- (a) What is the distance (in m) between the fence post and the street sign? [3]
(b) What is the angle (in degrees) between the surveyor and the fence post, as measured from the position of the street sign? [2]

- 1.2 In biology, a *vector* is a disease-carrying organism, such as a mosquito or a bat, that is itself not infected with the disease. The West Nile disease is spread primarily by migratory birds. The figure below shows the spread of the West Nile disease causing virus across the continental United States from 1999 to 2004.

The West Nile Virus was first detected in New York in 1999. Two years later, the virus had spread to Miami, a distance of 1,100 miles at a direction of 7.00° west of south from New York. In 2002, the virus was found in Minneapolis, 1,500 miles at a direction 37.0° west of north from Miami. Finally, the virus arrived in San Diego in 2004, a distance of 1,550 miles at a direction 25.0° south of west.

- (a) What is the net displacement (in mi) of the virus from 1999 to 2004? (Give the direction in degrees south of west, rounded to at least three significant figures.) [6]
(b) What is the average speed (in mi/h) of the spread of the virus from New York to San Diego during this time? [2]

- 1.3 In an old mining report, you read that a sample of limestone had a density of 156 pound per cubic foot. Given that 1 pound = 453.592 g, 1 foot = 12 inch, and 1 inch = 25.4 mm, find the density of the sample in kg/m^3 . You must show all the appropriate conversion factors and express your answer in scientific notation to the correct number of significant digits. [4]

Question 2 [11]

A hotel elevator is moving downward in a multi-story building at a constant speed of 5.40 m/s. Exactly 3.21 s after the top of the elevator car passes a bolt loosely attached to the wall of the elevator shaft, the bolt falls from rest.

- (a) Show that if one ignores, for the time being, the numerical values in the statement above, one could develop an analytic / symbolic expression for the time it takes the bolt to hit the top of the still-descending elevator from using basic kinematic equations. (Do not include any numeric values in this expression.) [6]
- (b) At what time (in s) does the bolt hit the top of the still-descending elevator? (Assume the bolt is dropped at $t = 0$ s.) [3]
- (c) Estimate the number of floors through which the bolt can fall if the elevator reaches the ground floor before the bolt hits the top of the elevator. (Assume 1 floor = 3 m.) [2]

Question 3 [15]

It is not possible to see very small objects, such as viruses, using an ordinary light microscope. An electron microscope can view such objects using an electron beam instead of a light beam. Electron microscopy has proved invaluable for investigations of viruses, cell membranes and subcellular structures, bacterial surfaces, visual receptors, chloroplasts, and the contractile properties of muscles. The "lenses" of an electron microscope consist of electric and magnetic fields that control the electron beam.

As an example of the manipulation of an electron beam, consider an electron traveling away from the origin along the x axis in the xy plane with initial velocity $\vec{v}_i = v_1 \hat{i}$. As it passes through the region $x = 0$ to $x = d$, the electron experiences acceleration $\vec{a} = a_x \hat{i} + a_y \hat{j}$, where a_x and a_y are constants. (**NOTE:** You do not need to take gravity into account for this problem.)

- (a) Develop an analytic / symbolic expression for the position of the electron. Do not include any numerical values in this expression. [7]
- (b) For the case $v_1 = 1.67 \times 10^7$ m/s, $a_x = 8.50 \times 10^{14}$ m/s², and $a_y = 1.43 \times 10^{15}$ m/s², determine the following, when the electron has an x position, $x_f = d = 0.0100$ m:
 - (i) the position vector of the electron [2]
 - (ii) the velocity vector of the electron [4]
 - (iii) the speed of the electron [1]
 - (iv) the direction of travel of the electron (i.e. the angle between its velocity and the x axis) [1]

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